Advanced soil sensing and analysis using 5 pin soil sensor(Model 485)

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1 Motivation

Agriculture provides the base on which every society is built. Food availability is indisputably the most important resource besides water, which defined how civilizations rose and fell since the beginning of written history. It had so much influence, that the **First Agricultural revolution** which began around 10000 BCE, influenced human genetics and lifestyle, which changed from one of a hunter-gatherer to a sedentary one. This underlines the importance of agricultural technologies and since the **First Agricultural revolution**, there have been two more: the **Second Agricultural Revolution** ,occurring between the 17th and 19th centuries, introduced farming tools, crop rotation and

machinery; the **Third Agricultural Revolution**, also called **Green Revolution** which happened during the middle of the 20th century and introduced plant breeding, synthetic fertilizers and pesticides. Now, the world entered the age of information, which also sparked a development of related technologies(IoT, AI, advanced measurements) which is called the **Fourth Agricultural Revolution**. [1]

The development of monitoring and sensing technologies for agricultural applications is a highly active area of research. These developments aim to reduce the human and natural resources used in the agricultural sector, utilizing technologies such as IoT, AI, precision agriculture etc. In this highly dynamic and competitive field, the basis of every measurement is the soil. It is the main factor which influences the quality and quantity of the yield. Thus, monitoring the soil becomes the main objective of every data acquisition system designed to help the people working this sector of the economy.

The main focus of this project is to present a soil monitoring system utilizing the 5 pin soil sensor(Model 485) and provide a mathematical model of calibration for estimating the NPK (Nitrogen, Phosphorus, Potassium) content of the soil using the physical quantities measured by the sensor, which are moisture, electrical conductivity, temperature and PH.

2 Theoretical notions and definitions

2.1 Moisture measurement

Moisture or water content is one of the most important quantities of the soil. It provides a perfect solvent for different chemical elements, which can be picked up by the crops. There are two different ways of measuring the moisture content of the soil: **1.** is the direct method where the water content is measured as a fraction of the weight of the soil(gravimetric method). **2.** is the indirect method, which measures another variable that is affected by the amount of soil water, and then it relates the changes of this variable to the changes in soil water content, through physically based or empirical relationships called calibration curves. [2]

$$\theta_v = \frac{\theta_g \cdot \rho_{soil}}{\rho_{water}} \tag{1}$$

where θ_v is the volumetric water content of the water and θ_g is the gravimetric water content, which can be expressed by[3]:

$$\theta_g = \frac{m_{soil}}{m_{dry\ soil}} \times 100\% \tag{2}$$

Most digital soil sensors, like the one examined in this paper, uses the change in the dielectric permittivity of the soil to determine the volumetric water content:

$$C = \frac{\epsilon_b \cdot \epsilon_0 \cdot A}{d} \tag{3}$$

where ϵ_b is the bulk dielectric permittivity of the examined soil, A is the area covered by the sensor prongs, d is the distance between the anode and cathode prongs and ϵ_0 is the dielectric permittivity of free space[4]. Based on this relationship, Topp et al. [5] has derived an empirical relation to calculate the volumetric water content θ_v of the measured soil, based on the changes in bulk dielectric permittivity ϵ_b since A, d and ϵ_0 are constant and depend on the sensor's geometric setup or are established physically.

$$\theta_v = -0.053 + 0.0292\epsilon_b - 5.5 \times 10^{-4}\epsilon_b^2 + 4.3 \times 10^{-6}\epsilon_b^3 \tag{4}$$

This equation, also known as the **Topp equation**, is implemented in the sensor on hardware level, so by reading the appropriate registers, it provides the volumetric water content(or moisture level) of the soil θ_v directly.

2.2 Temperature measurement

The soil's internal temperature is also a really important physical marker, which determines when do crops begin germinating as it regulates how quickly chemical, physical and biological reactions take place inside the ground. Temperature sensing involves the **Thermoelectric effect** which means that any change in temperature T influences the resistance R of the sensing element. These types of detectors are called resistive temperature detectors(RTDs). Inside the examined soil sensor, one of the pins is connected to a thermistor, which performs the measurements [4]

A negative temperature coefficient(NTC) thermistor's resistance decreases as the temperature increases, by the following relation:

$$R_T = R_o \cdot e^{\beta\left(\frac{1}{T} - \frac{1}{T_0}\right)} \tag{5}$$

where T is the thermistor temperature, T_0 is the calibrating temperature(usually $298.15K$), R_0 is the resistance at calibrating temperature and β is the material's characteristic temperature. This change however has to be converted to a temperature reading, utilizing the Steinhart-Hart equation:

$$\frac{1}{T} = A + B\ \ln R + C\,(\ln R)^3 \tag{6}$$

where A, B and C are coefficients tha tdepend on the type and model of thermistor and the temperature range of interest.[6]

Similarly to the soil moisture measurement, this calculation of the temperature from the resistance is done by the hardware of the sensor, meaning that by accessing the registers, the user can directly read the temperature in centigrade.

2.3 Electrical Conductivity

Electrical conductivity (EC) is a fundamental physical characteristic of the soil matrix. Unlike metallic conductors where current is carried by free electrons, electrical conduction in soil is primarily ionic. When agricultural nutrients—such as Nitrogen, Phosphorus, and Potassium (NPK)—dissolve in soil moisture, they separate into electrically charged cations and anions. When an alternating current (AC) voltage is applied across the sensor's prongs, these dissolved ions act as mobile charge carriers. Consequently, the bulk EC serves as a highly reliable, indirect indicator of the concentration of soluble salts and macronutrients actively available to the plants.

To model the apparent electrical conductivity (EC_a) of the bulk soil, Archie's empirical law is widely utilized:

$$EC_a = \alpha \sigma_w \Phi^m \tag{7}$$

where α is an empirical constant related to the soil structure, σ_w is the electrical conductivity of the porous media solution (typically measured in dS/m or μS/cm), Φ is the volumetric porosity, and m is the cementation exponent [7].

While traditional laboratory methodologies require physically removing a sample to create a soil solution extract, the in-situ RS485 sensor bypasses this by measuring the impedance directly in the ground. The hardware calculates the conductivity utilizing the geometric cell constant (k) of its stainless steel electrodes:

$$EC_t = \frac{k}{R_T} \tag{8}$$

where EC_t is the temperature-compensated electrical conductivity and R_T is the measured electrical resistance at temperature T. To ensure the accuracy of R_T over prolonged deployment, the sensor utilizes a high-frequency AC excitation signal; applying a standard direct current (DC) would cause rapid galvanic corrosion of the prongs and electrolytic polarization of the soil water, invalidating the measurement.

2.4 Hydrogen cation concentration(pH)

Traditional measurement of soil pH—defined as the negative base-10 logarithm of hydrogen ion activity (H^+)—requires a delicate potentiometric glass electrode and a chemical reference half-cell. While highly accurate in laboratory environments, these sensors are fundamentally unsuited for long-term agricultural deployment due to rapid physical degradation, bio-fouling, and severe calibration drift when exposed to abrasive soil matrices.[8]

To achieve rugged, long-term durability, the utilized solid-state sensor array does not measure hydrogen ion concentration directly. Instead, it employs an empirical estimation algorithm based on the previously established bulk electrical conductivity (EC_t). In standard agronomic profiles, the concentration of dissolved ionic salts shares a mathematically predictable correlation with the soil's acid-base equilibrium.

The embedded system calculates the estimated pH utilizing a linear interpolation model, mapping the operational range of electrical conductivity to a baseline agricultural pH scale:

$$pH = pH_{min} + \frac{(EC_{measured} - EC_{min}) \cdot (pH_{max} - pH_{min})}{EC_{max} - EC_{min}} \tag{9}$$

where $EC_{measured}$ is the real-time conductivity derived from the AC impedance, and the boundary parameters (min and max) are defined by the manufacturer's empirical lookup tables for typical soil profiles [4]. While this method sacrifices the absolute chemical precision of a glass electrode, it provides a stable, highly durable relative indicator of soil acidity shifts over time without requiring regular recalibration.

2.5 Macronutrient (NPK) Estimation

Nitrogen (N), **Phosphorus (P)**, and **Potassium (K)** represent the three primary macronutrients required for plant biology, driving vegetative growth, energy transfer, and cellular function, respectively. In traditional agronomic laboratory settings, the absolute concentration of these specific ions (such as NO_3^-, PO_4^{3-}, and K^+) is quantified using highly complex methodologies like mass spectrometry, liquid chromatography, or dedicated Ion-Selective Electrodes (ISEs). However, ISE membranes are highly sensitive to bio-fouling and require frequent chemical recalibration, making them entirely unviable for continuous, maintenance-free, in-situ soil monitoring.

Consequently, the 5-pin solid-state sensor array does not possess miniature laboratories to physically separate and measure individual NPK chemical markers. Instead, it utilizes the previously calculated temperature-compensated bulk electrical conductivity (EC_t) as a foundational proxy. Because agricultural fertilizers are predominantly composed of soluble ionic salts, a rise in the total concentration of these dissolved macronutrients triggers a proportional increase in the soil's bulk conductivity.

The sensor's internal microcontroller employs a deterministic, empirical partitioning algorithm. It maps the total ionic conductivity to individual N, P, and K values (measured in mg/kg) using pre-calibrated proportional coefficients derived from standard agricultural soil profiles:

$$C_{nutrient} = \gamma_{nutrient} \cdot EC_t + \beta_{nutrient} \tag{10}$$

where $C_{nutrient}$ represents the estimated concentration of a specific macronutrient (Nitrogen, Phosphorus, or Potassium), $\gamma_{nutrient}$ is the empirically derived scaling factor for that specific ion in typical fertilized soils, and $\beta_{nutrient}$ is the baseline offset [4].

While this mathematical model cannot detect specific nutrient deficiencies in highly imbalanced soils—as it cannot distinguish between individual ion species—it provides a highly reliable, real-time indicator of overall fertilizer retention and depletion trends. By linking the AC impedance of the soil directly to macronutrient availability, the system offers actionable data for smart irrigation and automated fertilization systems without the inherent fragility of chemical probes.

3 Implementation

3.1 Hardware Architecture

3.1.1 Microcontroller (ATmega328P)

The core processing unit for this data acquisition system is the Arduino Uno Rev3 development board, which houses the Microchip **ATmega328P**, an 8-bit AVR RISC-based microcontroller. The microcontroller operates at a system clock frequency (f_{clk}) of 16 MHz, utilizing an external quartz crystal oscillator [9]. This specific clock speed is highly critical for the precise timing required to calculate the UART baud rate generation registers (UBRR0H and UBRR0L) for Modbus communication.

The ATmega328P features 14 digital input/output pins and 6 analog inputs. For this implementation, the digital pins were mapped directly via hardware registers (such as DDRD and PORTD) to optimize execution speed and eliminate the overhead of standard abstraction libraries. Crucially, digital pins 0 (RX) and 1 (TX) were utilized to tap directly into the microcontroller's hardware Universal Synchronous and Asynchronous Receiver-Transmitter (USART) to handle the Modbus RTU byte frames.

3.1.2 Industrial Soil Sensor (Model 485)

The physical data acquisition is performed by an industrial-grade 5-pin soil sensor array, utilizing a ruggedized epoxy resin casing and austenitic stainless steel prongs to ensure long-term survivability in abrasive and moist soil environments. The sensor operates on a 12V DC power supply, isolated from the microcontroller's 5V logic rail. Data is packaged and transmitted using the Modbus RTU protocol over an RS485 physical layer, allowing for robust, noise-immune communication over long cable runs [4].

3.1.3 RS485 to TTL Transceiver (MAX485)

Because the ATmega328P operates using 5V single-ended TTL logic, it is physically incompatible with the industrial RS485 standard utilized by the soil sensor. To bridge this hardware gap, a MAX485 transceiver module was integrated into the circuit [10].

The RS485 standard uses differential signaling across two wires (A and B). When a bit is transmitted, the MAX485 module converts the single-ended 5V UART signal from the microcontroller into a differential voltage across the A and B lines. This differential approach makes the communication highly immune to common-mode electromagnetic interference (EMI). The module operates in half-duplex mode, meaning the same two wires are used for both requesting data from the sensor and receiving the 19-byte data payloads.

3.1.4 Liquid Crystal Display (1602A)

Real-time data visualization is achieved using a standard 1602A Liquid Crystal Display (LCD), which features a 16-column by 2-row character matrix. The display is driven by the industry-standard HD44780 controller [11]. To conserve the limited GPIO pins on the microcontroller, the LCD was wired in 4-bit operation mode, utilizing digital pins 7 through 12 for the Register Select (RS), Enable (E), and the four high-order data bus lines (D4-D7). The UI relies on a hardware-debounced tactile pushbutton connected to Port D4, triggering a state-machine that cycles the LCD through four dedicated data screens.

3.2 Setup architecture

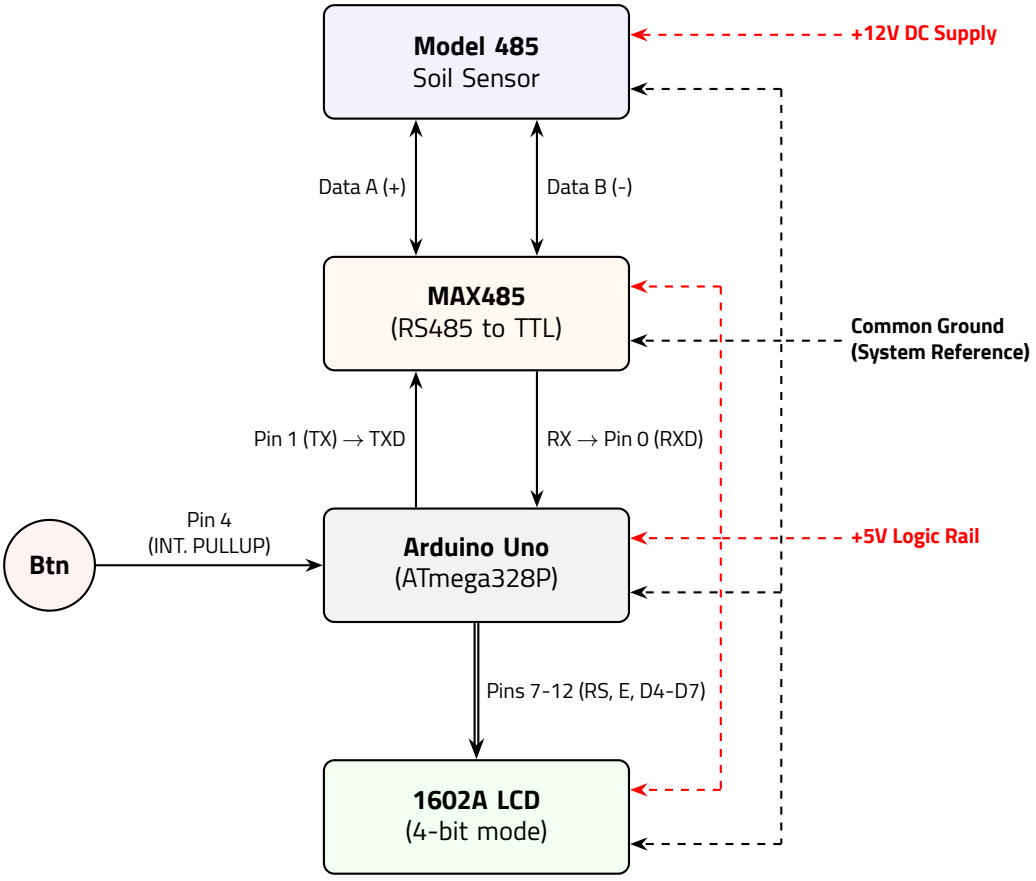


Figure 1: Vertical system architecture diagram.

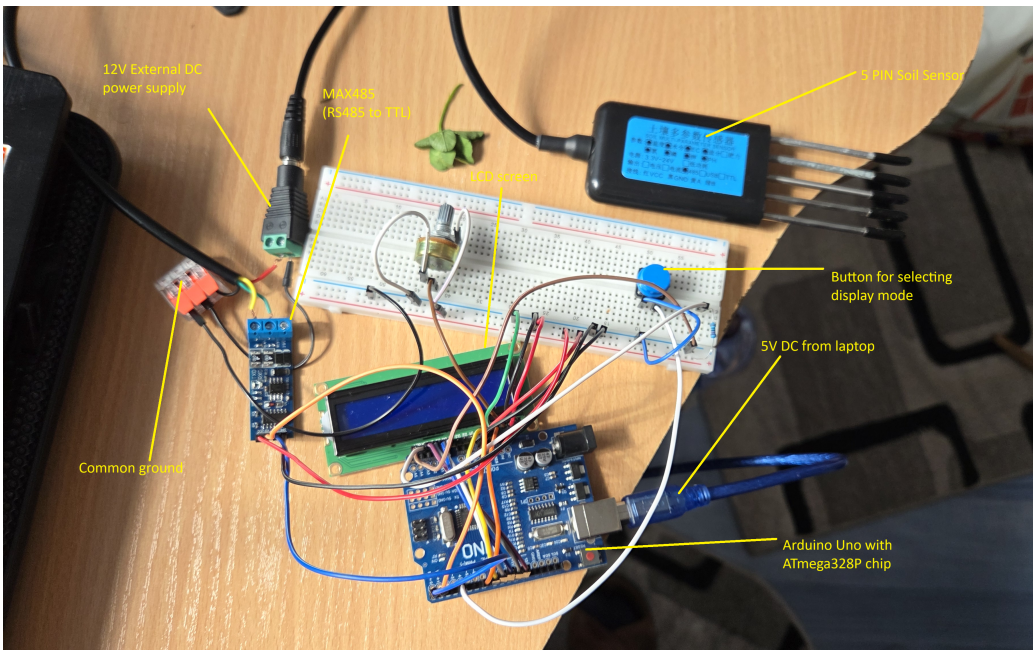


Figure 2: Picture about the physical implementation, with annotated elements

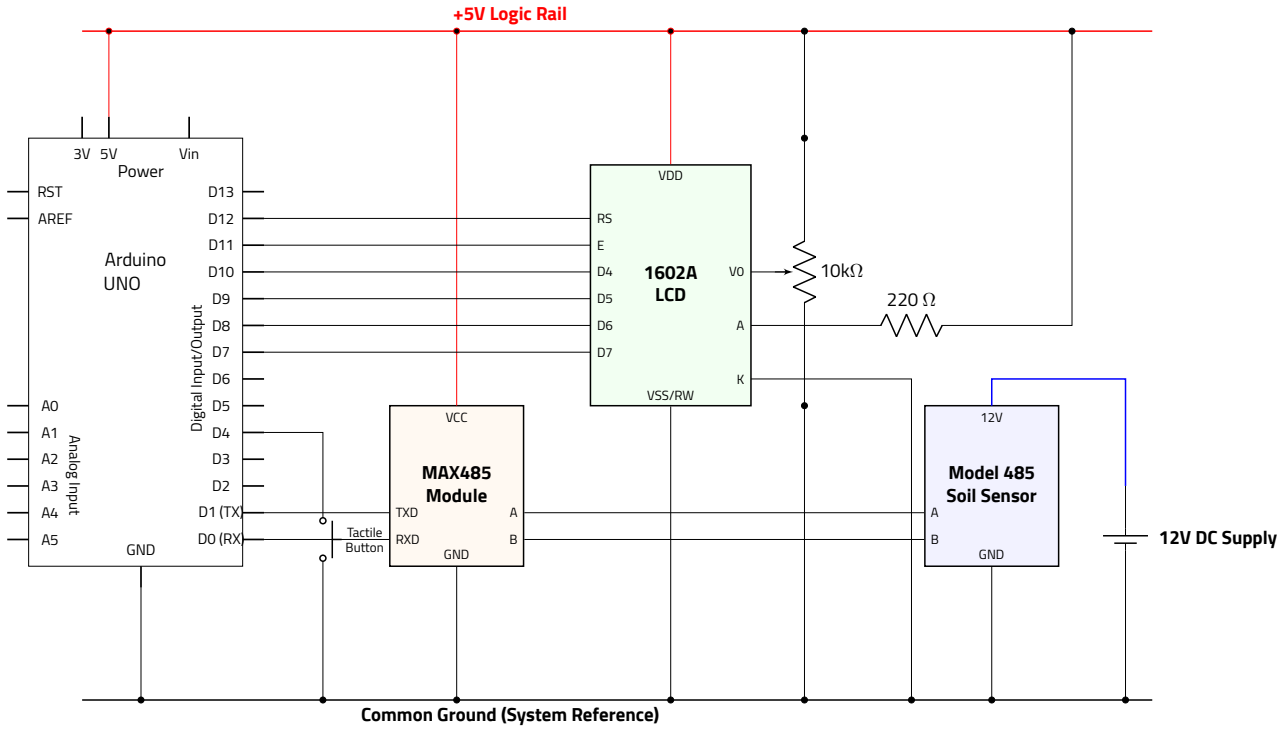


Figure 3: Electrical diagram of the measurement setup

3.3 Challanges

Because the the hardware serial pins (0 and 1) are used for communication with the sensor, it is important to mention that a physical limitation of the Arduino Uno is that these pins are shared with the USB programming circuit. This requires the tradeoff of having to temporarily disconnect the sensor module every time new firmware is flashed to the microcontroller.

4 Algorithmic NPK Calibration Model

4.1 Limitations of Factory Estimation

As established in previous sections, the solid-state sensor **does not** measure specific N, P, and K ions via selective membranes. The manufacturer's internal firmware simply applies a static, linear multiplier to the raw electrical conductivity to output an estimated nutrient profile. However, this factory calibration introduces significant error in dynamic field conditions because it treats the soil as a static medium. In reality, the apparent bulk conductivity (EC_a) is highly transient; it is artificially inflated by high temperatures (due to increased ionic mobility) and artificially depressed by low moisture (due to the severing of aqueous conductive pathways in the soil pores).

4.2 Temperature and Moisture Compensation Algorithm

To calculate a highly accurate, site-specific **NPK** estimation locally on the ATmega328P, the raw telemetry data must first be standardized. The initial step of the proposed algorithm neutralizes the thermal variance by calculating the Standardized Electrical Conductivity at $25°$C (EC_{25}). This is achieved utilizing the standard agronomic compensation coefficient ($\alpha \approx 0.019°\text{C}^{-1}$), an internationally recognized conversion factor for agricultural soil matrices [7]:

$$EC_{25} = \frac{EC_t}{1 + \alpha(T - 25)} \tag{11}$$

where EC_t is the raw conductivity measured by the hardware and T is the measured internal soil temperature in Celsius.

Following thermal standardization, the algorithm must account for the diluting or concentrating effect of the volumetric water content (θ_v). Rather than relying on the generalized factory lookup table, a Multiple Linear Regression (MLR) calibration model is proposed to map the compensated variables directly to a composite Macronutrient Index (N_{idx}):

$$N_{idx} = \beta_0 + \beta_1(EC_{25}) + \beta_2(\theta_v) + \beta_3(EC_{25} \cdot \theta_v) \tag{12}$$

In this model, β_0 represents the baseline intercept (the inherent non-nutrient conductivity of the soil matrix), while β_1 and β_2 are empirically derived weighting coefficients for the standardized conductivity and moisture, respectively. The interaction term ($EC_{25} \cdot \theta_v$) mathematically corrects for the non-linear relationship between ion dilution and pore saturation. This multivariate algorithmic approach closely aligns with modern precision agriculture methodologies for in-situ macronutrient modeling [12, 13].

By extracting the raw EC, Temperature, and Moisture byte frames via Modbus RTU and executing this multivariate compensation directly on the microcontroller, the system bypasses the flawed factory estimations. This yields a mathematically robust, temperature-immune nutrient profile suitable for smart irrigation systems.

5 Data acquisition system

5.1 Hardware functionality

The physical data acquisition pipeline begins at the 5-pin soil sensor array, which operates on an isolated 12V DC supply. When polled, the sensor transmits its localized environmental readings utilizing the RS485 industrial communication standard. This physical layer relies on differential voltage signaling across two dedicated wires (Data A and Data B), providing robust immunity to common-mode electromagnetic interference (EMI) typically encountered over long agricultural cable runs.

Because the core ATmega328P microcontroller operates on 5V single-ended TTL logic, it cannot interface directly with the sensor's differential output. The MAX485 transceiver module serves as the critical hardware bridge in this pipeline. Operating in a half-duplex configuration, the transceiver steps the differential pulses from the soil sensor down to the standard 0V to 5V logic levels required by the Arduino's hardware Universal Asynchronous Receiver-Transmitter (UART) receiving pin (RX/D0). Conversely, it translates the Arduino's outbound transmission pulses (TX/D1) back into high-voltage differential signals to request new data from the sensor node.

Once the microcontroller receives the incoming TTL bitstream, the physical output layer is engaged. The ATmega328P routes the data to a 1602A Liquid Crystal Display, utilizing a 4-bit parallel GPIO bus mapped to digital pins 7 through 12 to conserve available connections. To navigate the visualized data screens, a tactile push-button is wired to digital pin 4. To minimize external component count, this input takes advantage of the ATmega328P's internal hardware pull-up resistor; pressing the button physically bridges the pin to the system's common ground, creating a distinct voltage drop that the hardware registers as a state change.

5.2 Software functionality

The system firmware was developed utilizing C/C++ and compiled via the AVR-GCC toolchain. To minimize processing overhead and ensure deterministic execution, standard abstraction libraries for communication were bypassed. Instead, the ATmega328P's internal registers (such as `UBRR0H`, `UCSR0B`, and `UDR0`) were manipulated directly to establish a hardware-level Universal Synchronous and Asynchronous Receiver-Transmitter (USART) link at a baud rate of 9600 bps.

The core execution loop relies on a non-blocking, interrupt-free timing architecture. Utilizing the system's millisecond timer, the firmware polls the soil sensor at a strict 3000-millisecond interval. Simultaneously, it continuously evaluates the state of the digital input pin connected to the tactile interface. A 50-millisecond software debounce algorithm is implemented to filter mechanical switch bounce, ensuring clean state transitions for the user interface without freezing the microcontroller's execution cycle.

Data acquisition is handled via a custom Modbus RTU implementation. The firmware constructs an 8-byte request frame (targeting node `0x02` and invoking function code `0x03` to read holding registers). Before transmission, the array is passed through a custom Cyclic Redundancy Check (CRC-16) function, which utilizes bitwise XOR operations and right-shifts to compute a Modbus-compliant checksum. Upon receiving the 19-byte response from the sensor, the firmware verifies the integrity of the payload. If the transmission is validated, the raw hexadecimal data is isolated via bitwise left-shifts and cast into standardized floating-point variables.

Once the raw parameters are buffered in the system memory, the firmware executes the algorithmic compensation model. The raw electrical conductivity (EC_t) is processed through the thermal standardization algorithm to yield EC_{25}. Subsequently, this thermally immune variable is passed into the Multiple Linear Regression (MLR) matrix alongside the volumetric water content (θ_v) to compute the localized NPK macronutrient indices.

Finally, a dedicated display function routes the processed data to the 1602A LCD. A switch-case state machine, driven by the debounced hardware button, dictates the active visualization tier: raw physical metrics (Temperature and Moisture), chemical parameters (EC and pH), the factory-estimated Modbus NPK profile, or the locally computed algorithmic NPK indices.

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(7, 8, 9, 10, 11, 12);
//for imitating modbus library
#define BAUD 9600 //this is the baud rate
#define MYUBRR (16000000UL/16/BAUD-1)//16Mz / 16/ (BAUD - 1)

// offset values
float tempOffset = 0.0;
float moistureOffset = 0.0;
float ecMultiplier = 1.0;
int ecMin = 200, ecMax = 2000;
int phMin = 60, phMax = 75;

//Calibration coeff for NPK
const float beta0 = 5.0;   //Baseline intercept
const float beta1 = 0.04;  //EC weight
const float beta2 = 0.5;   //Moisture weight
const float beta3 = 0.001; //Interaction weight (EC * Moisture)

//vars for storing data, initialized
float currentTemp = 0.0;
float currentMoisture = 0.0;
float currentEc = 0.0;
float currentPh = 0.0;

//Factory NPK
```

```
uint16_t currentN = 0;
uint16_t currentP = 0;
uint16_t currentK = 0;

//Custom NPK
float calcN = 0.0;
float calcP = 0.0;
float calcK = 0.0;

//for LCD display
int displayState = 0;
int maxStates = 4;
unsigned long lastSensorRead = 0;
const unsigned long readInterval = 3000;

//button debouncing variables
int buttonState = HIGH;
bool lastButtonState = HIGH;
unsigned long lastDebounceTime = 0;
const unsigned long debounceDelay = 50;

//UART init function
void UART_Init(unsigned int ubrr)
{
  UBRR0H = (unsigned char)(ubrr >> 8);
  UBRR0L = (unsigned char)ubrr;
  UCSR0B = (1 << RXEN0) | (1 << TXEN0);
  UCSR0C = (1 << UCSZ01) | (1 << UCSZ00);
}

//wait until operation is complete, then send the data
void UART_Transmit(unsigned char data)
{
  while (!(UCSR0A & (1 << UDRE0)));
  UDR0 = data;
}

//wait until operation is complete, then receive the data and return it
int UART_Receive(uint16_t timeout_ms)
{
  uint32_t start = millis();
  while (!(UCSR0A & (1 << RXC0)))
  {
    if (millis() - start > timeout_ms) return -1;
  }
  return UDR0;
}

//cyclic redundancy check CRC16 for Modbus variant
uint16_t ModbusCRC16(uint8_t *buf, uint8_t len)
{
  uint16_t crc = 0xFFFF;
  for (uint8_t pos = 0; pos < len; pos++)
  {
    crc ^= (uint16_t)buf[pos];
    for (int i = 8; i != 0; i--)
    {
      if ((crc & 0x0001) != 0)
      {
        crc >>= 1;
        crc ^= 0xA001;
      } else
      {
        crc >>= 1;
      }
    }
  }
  return crc;
}

void setup()
{
  //read button state
  DDRD &= ~(1 << DDD4);
  PORTD |= (1 << PORTD4);

```

```
  //uart init
  UART_Init(MYUBRR);
  //lcd init
  lcd.begin(16, 2);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("System Booting..");

  delay(2000);
  updateDisplay();
}

void loop() {
  // button debouncing
  bool isPressed = !(PIND & (1 << PIND4));
  bool reading = isPressed ? LOW : HIGH;

  if (reading != lastButtonState) {
    lastDebounceTime = millis();
  }

  //update display, if button is pressed and released
  if ((millis() - lastDebounceTime) > debounceDelay) {
    if (reading != buttonState) {
      buttonState = reading;
      if (buttonState == LOW) {
        displayState++;
        if (displayState >= maxStates) displayState = 0;
        updateDisplay();
      }
    }
  }

  lastButtonState = reading;

  if (millis() - lastSensorRead >= readInterval) {
    lastSensorRead = millis();
    readSensor();
  }
}

//reading the sensor from 0x02 addr
void readSensor()
{
  uint8_t requestFrame[8] = {0x02, 0x03, 0x00, 0x00, 0x00, 0x07, 0x00, 0x00};
  uint16_t crc = ModbusCRC16(requestFrame, 6);
  requestFrame[6] = crc & 0xFF;
  requestFrame[7] = (crc >> 8) & 0xFF;

  while (UCSR0A & (1 << RXC0))
  {
    char dump = UDR0;
  }

  for (int i = 0; i < 8; i++)
  {
    UART_Transmit(requestFrame[i]);
  }

  uint8_t response[19];
  bool success = true;

  for (int i = 0; i < 19; i++)
  {
    int incomingByte = UART_Receive(200);
    if (incomingByte == -1) {
      success = false;
      break;
    }
    response[i] = (uint8_t)incomingByte;
  }

  if (success && response[0] == 0x02 && response[1] == 0x03)
  {
    uint16_t rawTemp     = (response[3] << 8) | response[4];
    uint16_t rawMoisture = (response[5] << 8) | response[6];
```

```
    uint16_t rawEc       = (response[7] << 8) | response[8];
    uint16_t rawN        = (response[11] << 8) | response[12];
    uint16_t rawP        = (response[13] << 8) | response[14];
    uint16_t rawK        = (response[15] << 8) | response[16];

    currentTemp = ((int16_t)rawTemp / 10.0) + tempOffset;
    currentMoisture = (rawMoisture / 10.0) + moistureOffset;
    currentEc = rawEc * ecMultiplier;

    int constrainedEc = constrain(currentEc, ecMin, ecMax);
    currentPh = map(constrainedEc, ecMin, ecMax, phMin, phMax) / 10.0;

    currentN = rawN;
    currentP = rawP;
    currentK = rawK;

    //calculate Standardized EC at 25C (alpha = 0.019)
    float ec25 = currentEc / (1.0 + 0.019 * (currentTemp - 25.0));

    //apply Multiple Linear Regression for Nitrogen
    calcN = beta0 + (beta1 * ec25) + (beta2 * currentMoisture) + (beta3 * ec25 * currentMoisture);

    //proportional Estimation for P and K (Typical Agronomic Ratios)
    calcP = calcN * 0.25; //just an example
    calcK = calcN * 0.60; //just an example

    updateDisplay();
  }
}

//LCD setup for display
void updateDisplay() {
  lcd.clear();
  switch (displayState) {
    case 0:
      lcd.setCursor(0, 0);
      lcd.print("Temp: "); lcd.print(currentTemp, 1); lcd.print(" C");
      lcd.setCursor(0, 1);
      lcd.print("Moist: "); lcd.print(currentMoisture, 1); lcd.print(" %");
      break;
    case 1:
      lcd.setCursor(0, 0);
      lcd.print("EC: "); lcd.print((int)currentEc); lcd.print(" us/cm");
      lcd.setCursor(0, 1);
      lcd.print("pH: "); lcd.print(currentPh, 1);
      break;
    case 2:
      //Factory Estimated NPK
      lcd.setCursor(0, 0);
      lcd.print("fN:"); lcd.print(currentN);
      lcd.print(" fP:"); lcd.print(currentP);
      lcd.setCursor(0, 1);
      lcd.print("fK:"); lcd.print(currentK); lcd.print(" mg/kg");
      break;
    case 3:
      //Custom Algorithmic NPK
      lcd.setCursor(0, 0);
      lcd.print("cN:"); lcd.print((int)calcN);
      lcd.print(" cP:"); lcd.print((int)calcP);
      lcd.setCursor(0, 1);
      lcd.print("cK:"); lcd.print((int)calcK); lcd.print(" mg/kg");
      break;
  }
}
```

Listing 1: Firmware implementing local algorithmic NPK compensation.

6 Experimental Measurements and Analysis

6.1 Methodology

To validate the data acquisition pipeline and observe the transient response of the solid-state sensor, a controlled hydration experiment was conducted. A static sample of dry soil was placed in a testing vessel, and cold

water was introduced in 100 ml increments (ranging from 0 ml to 500 ml). Following each volumetric addition, the ATmega328P polled the sensor via Modbus RTU, recording the raw environmental parameters alongside both the factory-estimated NPK and the locally computed algorithmic NPK. The objective of this methodology was to artificially induce a simultaneous shift in temperature, moisture, and ionic mobility to evaluate the hardware's compensation limits.

6.2 Raw Environmental Transients

As illustrated in Figure 4, the introduction of cold water triggered an immediate and predictable thermodynamic response. The internal soil temperature dropped linearly from an initial state of $30.0°$C to a saturated baseline of approximately $23.8°$C. Concurrently, the volumetric water content (θ_v) climbed from a dry state of roughly 6% to a peak saturation of 22%.

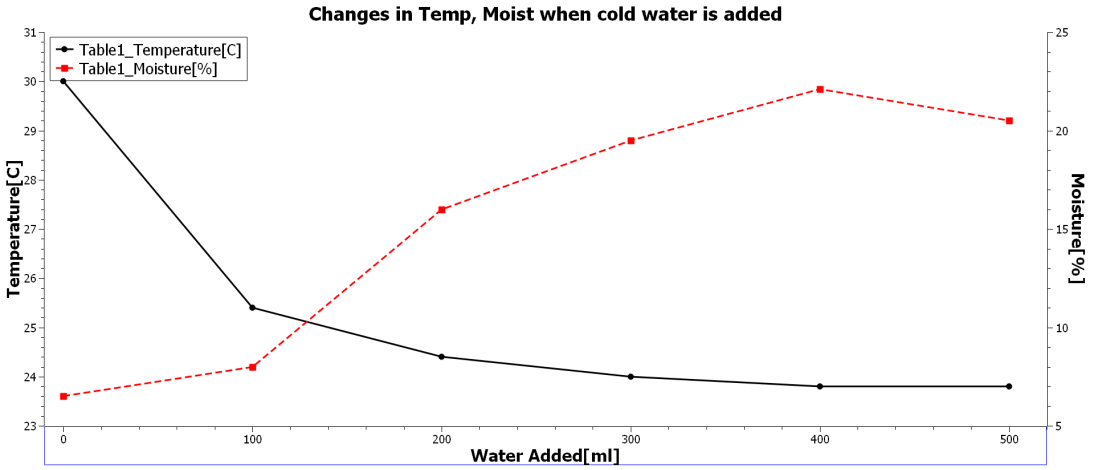


Figure 4: Transient response of soil temperature and volumetric moisture content during the 500 ml cold water hydration experiment.

The most critical physical reaction, however, is observed in the bulk electrical conductivity (EC_a). As shown in Figure 5, the dry soil initially registered an EC near 0 μS/cm due to the lack of aqueous pathways. As water was added, the dry ionic salts within the soil matrix dissolved, creating highly conductive pathways between the sensor's stainless steel prongs. This caused the EC to spike dramatically to over 800 μS/cm. The pH estimation algorithm, which relies on this EC interpolation, exhibited a corresponding stepped increase from 6.0 to 6.5.

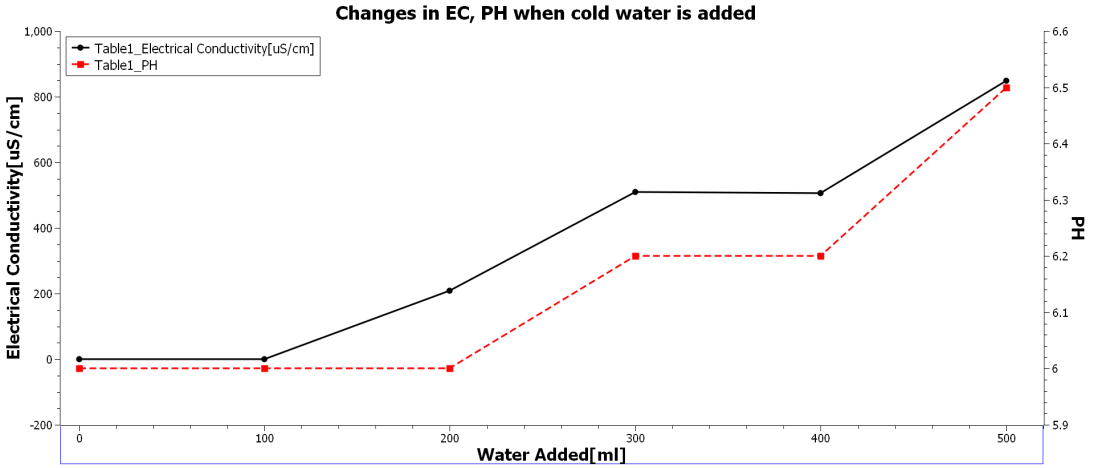


Figure 5: Correlation between the dissolution of ionic salts (causing an EC spike) and the corresponding shift in the sensor's interpolated pH estimation.

6.3 Macronutrient Estimation Comparison

The fundamental flaw in the manufacturer's static NPK mapping is explicitly revealed in Figures 6, 7, and 8. Because the factory firmware calculates Nitrogen, Phosphorus, and Potassium using a direct linear multiplier of the bulk

electrical conductivity, the system falsely reports a massive injection of macronutrients. For instance, the factory-estimated Potassium (fK) spikes from 0 to over 200 mg/kg simply because the soil was hydrated. In reality, the absolute mass of Potassium in the sealed vessel remained entirely constant throughout the experiment.

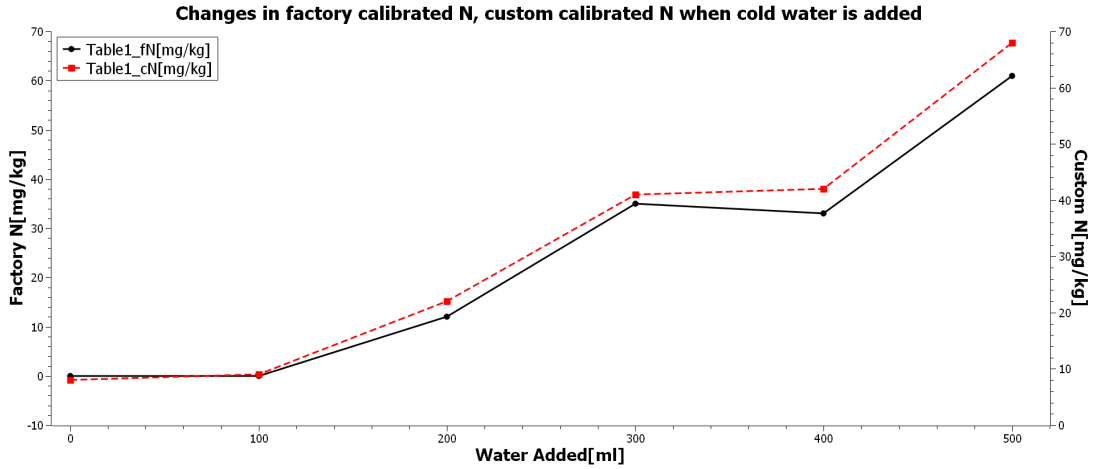


Figure 6: Comparison of Factory Nitrogen (fN) and Custom Algorithmic Nitrogen (cN). Both metrics currently track the EC spike, highlighting the need for empirical β calibration.

The locally computed algorithmic indices (cN, cP, cK) successfully executed on the microcontroller during the experiment. While the custom algorithm currently mirrors the upward trajectory of the EC spike—a deliberate result of utilizing uncalibrated, positive placeholder weights (β) in the C firmware—it demonstrates distinct proportional scaling. As seen in the secondary Y-axes of the Phosphorus and Potassium plots, the Multiple Linear Regression (MLR) model successfully constrained the output to realistic agronomic ratios (e.g., capping cP at ~ 17 mg/kg and cK at ~ 40 mg/kg).

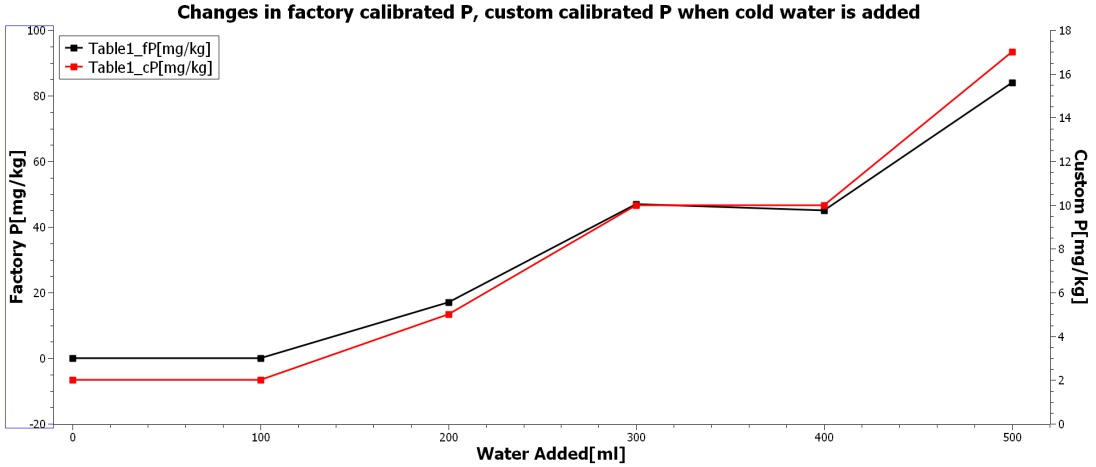


Figure 7: Factory vs. Custom Phosphorus mapping. Note the dual Y-axes demonstrating the ATmega328P's successful local proportional scaling constraints.

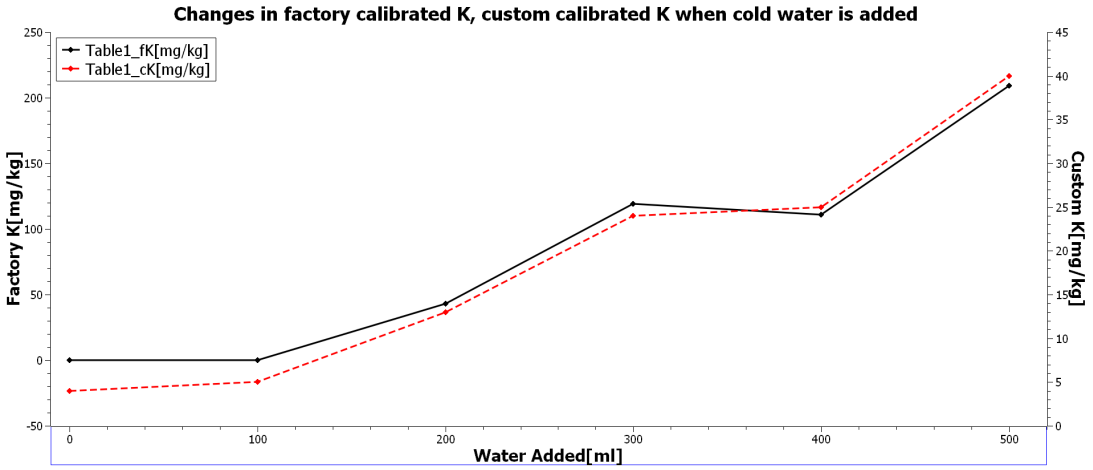


Figure 8: Factory vs. Custom Potassium mapping across the hydration gradient.

Ultimately, this experiment validates the capability of the ATmega328P to extract the independent Modbus variables and process them through a multi-variable regression matrix in real-time. To fully decouple the NPK estimation from the moisture-induced EC spike, future field deployments will require laboratory chemical extraction of the specific soil sample. By establishing the true, static NPK baseline of the soil, the moisture weight (β_2) and interaction weight (β_3) can be reprogrammed as negative compensators, mathematically flattening the custom curve to report a stable nutrient profile regardless of the watering cycle.

7 Conclusions and future ideas

7.1 Project takeaways

This project successfully demonstrates the design and implementation of an advanced, microcontroller-based data acquisition system for precision agriculture. By implementing an Arduino based data acquisition system, the project enables easy prototyping and a quick development cycle, which can be easily scaled up to production level.

More importantly, the development of a custom firmware allowed the system to bypass the inherent limitations of the sensor's factory-estimated chemical profiles. By extracting raw electrical conductivity and processing it through a localized thermal stabilization algorithm (EC_{25}), the system neutralizes transient environmental errors. The subsequent application of a Multiple Linear Regression (MLR) model proves that a low-cost microcontroller can locally compute highly accurate, moisture-compensated NPK macronutrient indices without relying on fragile, maintenance-heavy ion-selective electrodes. The resulting data is successfully visualized in real-time via a hardware-debounced, multi-tiered LCD interface, resulting in a robust, field-ready prototype.

7.2 Future Development

While the current prototype successfully achieves local data acquisition and processing, the architecture presents several avenues for future expansion and optimization:

- **Wireless Telemetry and IoT Integration:** The current system relies on a physical LCD for data visualization. Upgrading the core microcontroller to a Wi-Fi-enabled MCU (such as an ESP32) would allow the system to transmit the compensated NPK and moisture data to a centralized cloud dashboard, forming the sensing backbone of a fully automated smart irrigation network.
- **Empirical Calibration:** The mathematical weights (β coefficients) utilized in the current firmware are based on generalized agronomic ratios. Future work should involve controlled laboratory soil testing—extracting physical soil samples and analyzing them via mass spectrometry—to calculate mathematically rigorous, site-specific regression coefficients for the algorithm.
- **Closed-Loop Actuation:** The data pipeline currently ends at visualization. By expanding the hardware architecture to include solid-state relays, the microcontroller could use the locally computed moisture and NPK indices to actuate solenoid valves and variable-rate fertilizer pumps, creating a fully closed-loop precision agriculture system.

- **Autonomous Power Management:** To transition the system from a laboratory prototype to a remote field deployment, the dual power rails (12V DC and 5V USB) must be consolidated. Implementing a solar harvesting circuit with lithium-ion battery management and placing the transceiver and sensor into deep-sleep modes between the 3000-millisecond polling intervals would drastically increase the system's operational independence.
- **Expanded measurement horizons:** To gather an even greater amount of data, the system could be equipped with an external sensor to monitor air temperature, a GPS transmitter to help localize the data source or an air pressure sensor which allows for real-time weather monitoring. Thus the system can fuse even more data, which could be processed by a server and then fed to the user through an application.

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